ARCHITECT | BIM COORDINATOR | COMPUTATIONAL DESIGN

Riccardo Grillo



I coordinate multidisciplinary BIM projects for public infrastructure and automate workflows with Dynamo, Grasshopper and Rhino.Inside.Revit.

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01 — PROFILE

Architect specialised in Building Information Modeling (BIM), working as **BIM Coordinator** and **Computational Designer**. Experienced in coordinating multidisciplinary projects in the public infrastructure sector, particularly **railway infrastructure, bridges and viaducts**.

I coordinate the BIM team and support junior modellers, define **BIM standards and templates**, and automate modelling and information parametrization with **Dynamo** and **Grasshopper** (parametric design). Within the company I sit on the BIM Committee, the Cost Estimating Committee, the **AI Lab** and the **R&D** group on artificial intelligence applied to design, a topic I first explored in my master's thesis. My interests include digital survey, **Scan-to-BIM** and photogrammetry: I have produced BIM models of existing buildings and infrastructure from **point clouds**, including bridges and viaducts.

02 — PROFESSIONAL EXPERIENCE

Architect — BIM Coordinator and Structural BIM Specialist

May 2023 – Present

Masera Engineering Group · Turin, Italy

BIM DEVELOPMENT AND COORDINATION

- Architectural, structural and steelwork **BIM models** with **4D** (construction phasing and scheduling) and **5D** (cost estimating) logic; information parametrization and data management.
- **Clash detection in Navisworks**, **model checking** and parameter quality control, clash reports, check sheets and technical drawings.
- **Operational coordination** of BIM resources, providing technical support to 3–4 junior modellers / BIM Specialists.
- **Standards, procedures and best practices** for BIM Data Management; centralisation of in-house know-how.

COMPUTATIONAL DESIGN, AUTOMATION AND COST ESTIMATING

- Algorithms in **Grasshopper**, **Dynamo** and **Rhino.Inside.Revit** to automate modelling, parameter filling and drawing production.
- **Bills of quantities and cost estimates** in **TeamSystem STR**, linked to BIM models.

RESEARCH AND CORPORATE INNOVATION

- Member of the **BIM Committee**, **Cost Estimating Committee**, **AI Lab** and **R&D** group; testing of new software and BIM process digitalisation programmes.
- Designed and delivered an **internal seminar** on AI, prompt engineering and LLMs; co-author of contributions to **MEGazine** and technical articles in **FABER** and **Le Strade**.

KEY PROJECTS AND ACTIVITIES

01 **High-Speed Railway Line Salerno – Reggio Calabria (AV SA-RC).** Structural BIM Specialist and BIM Coordinator (approx. 16 months, team of 6; coordinated 4 BIM professionals). Contributed to over 40 BIM models (one per WBS) and over 140 technical drawings in Revit for the FOA noise barriers along approx. 33 km of line, with model coordination in Navisworks. Developed Dynamo and Grasshopper algorithms to automate modelling and information parametrization, with an estimated time saving of approx. 50%.

02 **Railway upgrades Messina – Catania (ME-CT) and Palermo – Catania (PA-CT).** ME-CT: produced approx. 20 BIM models of the FOA noise barriers and over 40 technical drawings in a team of 2 BIM resources. PA-CT: produced approx. 10 BIM models of technology yards and over 20 drawings as the sole BIM resource.

03 **Bridge maintenance – Turin – Bardonecchia railway line.** Structural BIM Specialist and BIM Coordinator (approx. 6 months, team of 3). Produced over 10 BIM models of bridges from point clouds (ReCap) for a maintenance-oriented digital twin; the project led to an article published in FABER.

04 **Artemio Franchi Stadium refurbishment, Florence.** Structural BIM Specialist and BIM Coordinator (approx. 4 months). Independently managed the entire BIM scope alongside one design engineer: modelled the stadium end stand (curva), handled model parametrization and export, and produced the technical drawings.

05 **Road bridges and viaducts.** BIM Coordinator, working with one design engineer. Produced 1 BIM model and approx. 5 technical drawings; composite steel-concrete and structural steel works, including the Steel Viaduct (Viadotto in Acciaio).

06 **BIM standardisation.** Created the reference Revit template (.rte), an operational guideline now in its third version and a model checklist; defined the graphic standards.

07 **Orte – Falconara railway line.** Activities related to the overpasses.

Architect — BIM Specialist

Jun 2022 – Feb 2023

Studio CABE — Cadacademy & Bim Engineering

- **BIM modelling** in Revit of the **UnipolSai Towers in Turin** from **point cloud surveys**: architectural and structural elements, phases, parametric families, filters and schedules.
- Information organisation for the export of **construction drawings**.

Freelance consultant · **2022 – Present**
Studio di ingegneria edile Mirenda (building engineering firm), Milazzo, Italy

Curricular internship · **Jul 2020 – Sep 2020**
Studio "Bartolo Doria architetto" (architecture firm), Milazzo, Italy

03 — SKILLS

- **Autodesk Revit** (Expert): Advanced architectural and structural modelling; templates and nested parametric families; schedules, drawings and design options; RVT, IFC, DWG, PDF.
- **Grasshopper** (Expert): Main Computational Design tool, connected to Revit via Rhino.Inside.Revit.
- **Autodesk Navisworks** (Advanced): Clash detection within and across models; multidisciplinary coordination; LC1–LC3 reports.
- **Dynamo** (Advanced): Connected to Revit to automate modelling and information parameters.
- **Rhinoceros 8** (Advanced): 3D modelling, mainly together with Grasshopper.
- **Graphic design and layout** (Advanced): Adobe Illustrator, Adobe InDesign.
- **Survey, photogrammetry and GIS** (Intermediate): Autodesk ReCap, RealityScan, Agisoft Metashape, CloudCompare, QGIS, RDF.
- **Rendering and visualisation** (Intermediate): Twinmotion, Enscape, Lumion.
- **Video and audio editing** (Intermediate): Adobe Premiere, DaVinci Resolve, OBS, Audacity.
- **AutoCAD** (Intermediate): Occasional use.
- **Productivity** (Intermediate): Microsoft Office (Word, Excel, PowerPoint), Google Sheets.
- **Programming** (Basic): Python, GhPython.
- **Cost estimating**: TeamSystem STR Vision CPM · quantity take-off linked to BIM models.
- **Cloud platforms**: Autodesk Construction Cloud (ACC), BIM 360.

AI AND DIGITAL PRODUCTIVITY TOOLS

Claude — reasoning, coding and digital solutions · **ChatGPT** — work organisation and recurring tasks · **Gemini** — in-depth research and analysis · **Gemini Notebook** — document and archive analysis · **Notion** — databases and knowledge management. Used to support research, organisation and automation.

04 — CERTIFICATIONS

openBIM	**buildingSMART Professional Certification — Foundation** — buildingSMART International · March 2026
English language	**IELTS** — Politecnico di Torino
Computational Design	**Learning Grasshopper** — Parametric modelling

05 — EDUCATION

2020 – 2023	**MSc in Architecture for Sustainable Design · 110/110** — Politecnico di Torino (Laurea Magistrale)
Sep 2021 – Feb 2022	**Erasmus exchange programme** — Universidad Politécnica de Madrid, ETSAM · Madrid, Spain (master's degree courses)
2017 – 2020	**BSc in Architecture** — Politecnico di Torino (Laurea)
2017	**Technical high school diploma — Tourism**

06 — PUBLICATIONS

Journal · 2026	**Integrazione tra BIM, LCA e Intelligenza Artificiale: verso una progettazione sostenibile e data-driven** **R. Grillo**, A. Surendiran · Le Strade, no. 1621, October 2026, pp. 31–37 · (Integrating BIM, LCA and AI)
Conference · 2026	**Playful Re-Design & Game Re-Design: Analysis, Ideation, Prototyping, and Game Strategy** U. Zich, A. Tomalini, F. Gibaja Gomez, **R. Grillo** · EDULEARN26 Proceedings, IATED · DOI 10.21125/edulearn.2026.1109
Journal · 2026	**BIM-Integrated Parametric Modelling of Existing Box Girder Bridges Using Point Cloud Data** A. Surendiran, **R. Grillo**, V. Boretti, M. Balestra, D. Masera · Int. Journal of Bridge Engineering, Management and Research · DOI 10.70465/ber.v3i1.75
MSc thesis · 2023	**Intelligenza artificiale in Architettura. Sviluppo generativo di soluzioni distributive attraverso l'uso di GANN** **R. Grillo** · Politecnico di Torino · Supervisors: M. Lo Turco, A. Tomalini, E. Pristeri · Webthesis

07 — TEACHING

A.Y. 2025/27	**Parametric and algorithmic modelling · Advanced parametric design** Academic tutor (approx. 40 students per course): lectures, reviews and exercises · Prof. M. Lo Turco; Prof. M. Caprani (taught in English)
A.Y. 2025/26	**PCTO outreach courses: Metaverse and 3D Rapid Prototyping for Heritage · Playful Re-Design** Collaborator (approx. 30 students): lesson organisation and student support · Prof. E. C. Giovannini; Prof. U. Zich
A.Y. 2024/25	**Architecture and computational design · Parametric and algorithmic modelling B** Academic tutor (approx. 40 students per course): lectures, reviews and exercises · Prof. M. Lo Turco; Prof. A. Tonin (taught in English)

08 — LANGUAGES AND INTERESTS

Languages	**Italian** native · **English** advanced · **Spanish** intermediate · **French** basic
Interests	Scan-to-BIM, Heritage BIM, BIM + VR, Project Management, photogrammetry and survey, GIS, AI automation and vibe coding, Visual Programming, sustainability protocols, cost estimating.
Hobbies	Beach volleyball, trekking, 3D printing.

I authorise the processing of my personal data under Regulation (EU) 2016/679 (GDPR) and Italian Legislative Decree 196/2003.